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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

BCSE204L- Design and Analysis of Algorithms

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Course Objective

1. To provide mathematical foundations for analysing the complexity of the algorithms
2. To impart the knowledge on various design strategies that can help in solving the real world problems effectively
3. To synthesize efficient algorithms in various engineering design situations



BCSE204L- Design and Analysis of Algorithms-

Course outcome

1. Apply the mathematical tools to analyse and derive the running time of the algorithms
2. Demonstrate the major algorithm design paradigms.
3. Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
4. Articulating Randomized Algorithms.
5. Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.



BCSE204L- Design and Analysis of Algorithms- Syllabus

Module:1	Design Paradigms: Greedy, Divide and Conquer Techniques	6 hours
Overview and Importance of Algorithms - Stages of algorithm development: Describing the problem, Identifying a suitable technique, Design of an algorithm, Derive Time Complexity, Proof of Correctness of the algorithm, Illustration of Design Stages - Greedy techniques: Fractional Knapsack Problem, and Huffman coding - Divide and Conquer: Maximum Subarray, Karatsuba faster integer multiplication algorithm.		
Module:2	Design Paradigms: Dynamic Programming, Backtracking and Branch & Bound Techniques	10 hours
Dynamic programming: Assembly Line Scheduling, Matrix Chain Multiplication, Longest Common Subsequence, 0-1 Knapsack, TSP- Backtracking: N-Queens problem, Subset Sum, Graph Coloring- Branch & Bound: LIFO-BB and FIFO BB methods: Job Selection problem, 0-1 Knapsack Problem		
Module:3	String Matching Algorithms	5 hours
Naïve String-matching Algorithms , KMP algorithm, Rabin-Karp Algorithm, Suffix Trees.		
Module:4	Graph Algorithms	6 hours
All pair shortest path: Bellman Ford Algorithm, Floyd-Warshall Algorithm - Network Flows: Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp, Push Re-label Algorithm – Application of Max Flow to maximum matching problem		
Module:5	Geometric Algorithms	4 hours
Line Segments: Properties, Intersection, sweeping lines - Convex Hull finding algorithms: Graham's Scan, Jarvis' March Algorithm.		
Module:6	Randomized algorithms	5 hours
Randomized quick sort - The hiring problem - Finding the global Minimum Cut.		
Module:7	Classes of Complexity and Approximation Algorithms	7 hours
The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover and Travelling salesman		
Module:8	Contemporary Issues	2 hours



BCSE204L- Design and Analysis of Algorithms- References

Text Books:

1. Thomas H. Cormen, C.E. Leiserson, R L. Rivest and C. Stein, Introduction to Algorithms, 2009, 3rd Edition, MIT Press.

Reference Books:

1. Jon Kleinberg, Éva Tardos ,Algorithm Design, Pearson education, 2014.
2. Rajeev Motwani, Prabhakar Raghavan; Randomized Algorithms, Cambridge University Press, 1995 (Online Print – 2013)
3. Ravindra K.Ahuja, Thomas L.Magnanti and James B.Orlin, “ Network Flows: Theory, Algorithms and Applications”, Pearson Education, 2014.



Module 3: String Matching Algorithms(5 Hours)

- **Naive String Matching Algorithm**
- KMP–Knuth Morris Pratt Algorithm
- Rabin- Karp Algorithm
- Suffix Trees



String Matching

- Text-editing programs frequently need to find all occurrences of a pattern in the text.
- Typically, the text is a document being edited, and the pattern searched for is a particular word supplied by the user.
- Efficient algorithms for this problem called **“String Matching”**
- **Example:**
 - Search for particular patterns in DNA sequences
 - Internet search engines – find Web pages relevant to queries.



Brute Force-Complexity

- Given a pattern “M” characters in length, and a text “N” characters in length...
- Worst case: compares pattern to each substring of text of length M. For example, M=5.

1) *AAAA*AAAAAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAH 5 comparisons made

2) *AAAA*AAAAAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAH 5 comparisons made

3) *AAAA*AAAAAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAH 5 comparisons made

4) *AAAA*AAAAAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAH 5 comparisons made

5) *AAAA*AAAAAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAH 5 comparisons made

....
N) AAAAAAAAAAAAAAAAAAAAAAAAAAAAAA*AAAAH*
5 comparisons made *AAAAH*

Total number of comparisons: $M(N-M+1)$

Worst case time complexity: $O(MN)$



Brute Force-Complexity(cont.)

- Given a pattern M characters in length, and a text N characters in length...
- Best case if pattern found: Finds pattern in first M positions of text. For example, M=5.

1) **AAAAA**AAAAAAAAAAAAAAAAAAAAAAAAAAH
AAAAA 5 comparisons made

Total number of comparisons: M
Best case time complexity: $O(M)$





Alg PatternMatching Brute Force (S,P)

{

LS=S.len

LP=P.len

MAX=LS-LP+1

for(i=1 to max)

{

Flag= True; // boolean variable

for(j=1;j<=LP && Flag ==True;j++)

{

if P[j] ~~=~~ S[j+i-1]

Flag= False;

} *// End of inner loop j*

if (flag==True)

return i;

} *// End of outer loop i*

return 0;

}

Basic idea to KMP Algorithm

TEXT

A B

PATTERN

A A A A A A B

Iteration 1

Mismatch

A A A A A A B

Iteration 2

In Naïve Based, We do right shift by one position to right and start checking the **PATTERN** from the first position..... **Why to the Beginning????**

Solution is : Just move 1 place before mismatch

